# Differentiable Simulation + Robot Arm + Policy Learning

CUSA Debulking: From Differentiable Sim to Real Robot

## 1. SAPO / SHAC: Short-Horizon Differentiable RL

SAPO (Soft Analytic Policy Optimization, ICLR 2025) and SHAC (Short-Horizon Actor-Critic, ICLR 2022) are the state-of-the-art methods for combining differentiable simulation with RL. They solve the long-rollout gradient explosion problem by only backpropagating through short windows (10-50 steps), using a learned critic to estimate long-term value.

### SAPO Algorithm Details

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SAPO (ICLR 2025, Rewarped Platform):

  Core idea: short-horizon analytic gradient + actor-critic RL

  For each training iteration:
    1. Sample parallel trajectories (N envs on GPU)
    2. Divide trajectory into short windows (10-50 steps)
    3. For each window:
       - Forward: run differentiable sim (Warp)
       - Backward: backprop gradient through sim
       - Policy gradient = analytic gradient from diff sim
       - Value target = multi-step reward + critic(last_state)
    4. Update actor with analytic policy gradient
    5. Update critic with TD learning (standard, no diff sim needed)

  Key differences from PPO/SAC:
    - Actor gradient comes from DIFF SIM (not reward sampling)
    - Critic is separate (trained via TD, not diff sim)
    - Maximum entropy formulation for exploration
    - 17x+ faster than PPO in wall-clock time
```

### SHAC vs SAPO vs PPO vs SAC

| Aspect | SAPO (2025) | SHAC (2022) | PPO | SAC |
|---|---|---|---|---|
| Gradient source | Diff sim (analytic) | Diff sim (analytic) | Reward sampling | Reward sampling |
| Backprop steps | 10-50 | 10-50 | 0 (no backprop) | 0 (no backprop) |
| Need diff sim? | Yes | Yes | No | No |
| Sample efficiency | Very high | Very high | Medium | High |
| Wall-clock speed | 17x vs PPO | 17x vs PPO | 1x (baseline) | ~2x vs PPO |
| Deformable? | Yes (Rewarped) | Mainly rigid | Any sim | Any sim |
| Stability | High (entropy) | High | Very high | Medium |
| On/off policy | On-policy | On-policy | On-policy | Off-policy |

## 2. Long-Rollout Gradient Problem & Solutions

Backpropagating through 100+ simulation steps causes: (1) GPU memory explosion, (2) gradient vanishing/exploding

(chain of matrix multiplications), (3) gradient discontinuity at contact events. This is the #1 open problem.

### Mitigation Strategies

| Strategy | Backprop Steps | Memory | Who Uses It |
|---|---|---|---|
| Truncated BPTT | 10-50 | Low | SAPO, SHAC (BEST for us) |
| Checkpointing | 100-500 | Medium | Warp, DiffTaichi |
| Implicit Diff (IFT) | Any | Low | Dojo, contact-implicit MPC |
| Adaptive Horizon | Variable | Medium | AHAC (auto-adjust) |
| Gradient Smoothing | Any | Low | Soft contact approx |

For our P(F) debulking sim: truncated BPTT (SAPO-style) is the best choice. Phase 4 verified 5-step gradient flow. SAPO uses 10-50 step windows. Our P(F) sigmoid is smooth, which helps gradient stability.

## 3. RL vs IL vs Diffusion: Which Method for Our Task?

| Method | Need Demo? | Need Diff Sim? | Sample Efficiency | Best For |
|---|---|---|---|---|
| SAPO (diff+RL) | No | Yes | Very high (17x PPO) | OUR BEST CHOICE |
| PPO (pure RL) | No | No | Low | Backup if diff fails |
| SAC (pure RL) | No | No | Medium | Off-policy backup |
| Diffusion Policy | Yes (50+ demos) | No | High (with demos) | Needs teleop setup |
| ACT | Yes | No | Medium | Real-time deploy |
| BC + diff refine | Few demos | Yes | High | Hybrid approach |

SAPO is our best choice because: (1) we have differentiable sim (P(F) + Warp), (2) no demonstration data needed, (3) 17x sample efficiency vs PPO, (4) the short-horizon approach matches our verified 5-step gradient capability.

## 4. Key Papers

**SAPO: Stabilizing RL in Differentiable Multiphysics Simulation**

*ICLR 2025*

Short-horizon analytic gradient + actor-critic. Handles rigid + articulated + deformable (elastic, plastic, fluid). Maximum entropy formulation. 17x faster than PPO. Our primary reference for training approach.

**SHAC: Short-Horizon Actor-Critic**

*ICLR 2022*

Foundation for SAPO. Truncated differentiable gradient + actor-critic updates. Parallel environment sampling. Gradient smoothing for non-smooth contact. Tested on locomotion + manipulation tasks.

**Stress-Guided RL for Gentle Manipulation of Deformable Objects**

*2025 | UFa*

Zero-shot sim-to-real for deformable manipulation. RL with stress-based reward. Real robot successfully manipulates tofu without damage. Domain randomization on material properties.

**HIL-SERL: Human-in-the-Loop Sample-Efficient RL**

*Science Ro*

Real robot RL with human corrections. Precise dexterous manipulation. Few-shot fine-tuning on real hardware. Practical sim-to-real bridge.

**SRT-H: Autonomous Surgical Task via Language-Conditioned IL**

*Science Ro*

Autonomous gallbladder removal on pig cadaver. Language-conditioned IL. 100% accuracy on standardized surgical tasks. da Vinci Research Kit.

**PlasticineLab: Soft-Body Manipulation Benchmark**

*ICLR 2021*

DiffTaichi backend. 100-200 step backprop works for plasticine. Trajectory optimization through differentiable MPM physics.

## 5. Sim-to-Real for Deformable Object Manipulation

### Three Approaches (Ranked by Maturity)

1. Domain Randomization (most mature): randomize tissue stiffness, damping, density, tool friction in sim. Train with randomized params. Deploy zero-shot. Success: tofu manipulation (xArm 7), tissue retraction (surgical robot).
2. Residual Sim-to-Real (emerging): train base policy in sim, learn residual correction from few real episodes. HIL-SERL (Science Robotics 2024): human-in-the-loop RL on real robot. Practical and sample-efficient.
3. Vision Domain Adaptation (surgical specific): GAN-based image translation from sim to real endoscopic images. Train vision policy on adapted images. 50% sim-to-real success for tissue retraction.

### Our Sim-to-Real Strategy

```
Step 1: Domain Randomization in Warp Sim
  - Tissue: stiffness [0.3, 1.0], damping [0.90, 0.99], density [800, 1200]
  - Tool: radius [0.8x, 1.2x], friction [0.1, 0.5]
  - Geometry: tumor shape variation (multiple .msh files)

Step 2: Train SAPO Policy
  - 16-32 parallel envs on single GPU
  - Short-horizon (10-50 steps) analytic gradient
  - P(F) geometry-aware reward: +volume - lambda*nerve_damage

Step 3: Zero-Shot Transfer to Real Robot
  - Franka Panda + CUSA-like tool on end-effector
  - Silicone 3-layer phantom (capsule + tumor + nerve)
  - RealSense depth camera for observation
  - If zero-shot fails: HIL fine-tuning (10-50 real episodes)
```

## 6. Differentiable Simulation Frameworks

| Framework | Backend | Deformable Support | Notes |
|---|---|---|---|
| Rewarped | Warp (CUDA) | Elastic, plastic, fluid | SAPO paper. Best for us |
| Genesis | DiffTaichi | MPM, FEM, PBD | 10-80x faster than others |
| Brax | JAX | Limited (rigid focus) | Massively parallel |
| Dojo | Julia | Limited | Hard contact + friction |
| DiffTaichi | Taichi | Elastic, cloth, fluid | 188x faster than TF |
| Newton | Warp+MuJoCo | Planned | NVIDIA 2025. Isaac Lab |

## 7. Our Complete Plan: P(F) Debulking + SAPO + Franka

```
=== System Architecture ===

  Isaac Lab (Franka Panda arm dynamics)
       |
       | end-effector position
       v
  Warp (P(F) Shape Matching tissue physics)
       |
       | damage, alpha, positions
       v
  SAPO Training (short-horizon diff grad + critic)
       |
       | trained policy
       v
  Real Franka + Silicone Phantom (sim-to-real transfer)


=== Training Details ===

  Observation (7D):
    - end-effector position (3D)
    - local P(F) mean around tool (1D)  <-- geometry-aware!
    - volume removed so far (1D)
    - nerve damage so far (1D)
    - distance to nearest nerve (1D)

  Action (3D):
    - end-effector velocity (dx, dy, dz)

  Reward:
    r = +delta_volume_removed
        - lambda * delta_nerve_damage
        - mu * ||action||^2  (energy penalty)
        + bonus * (volume > target)

  P(F) advantage:
    - nerve_damage = sum of damage on nerve-layer particles
    - P(F) automatically gives high value to nerve (sparse)
    - Gradient naturally guides tool AWAY from nerve


=== Timeline ===

  Week 1-2: Gym env wrapper + SAPO implementation
  Week 3-4: Training + hyperparameter tuning
  Week 5: Ablation (SAPO vs PPO vs SAC)
  Week 6: Visualization + paper figures
  Week 7: Sim-to-real (Franka + phantom)
  Week 8: Paper writing
```

## 8. Paper Framing & Venue Targets

### Title Options

(1) "Short-Horizon Differentiable Surgical Simulation with P(F) Geometry Sensing for Sample-Efficient Debulking Policy Learning"

(2) "P(F)-Guided Analytic Policy Gradient for Autonomous Vestibular Schwannoma Debulking"

(3) "Differentiable Geometry-Aware Reward Design for Surgical Debulking via SAPO"

### Key Claims

1. P(F) provides differentiable, geometry-aware damage sensing that naturally protects nerve tissue (vs hard thresholds in prior work)
2. Short-horizon autodiff through shape matching + P(F) sigmoid enables analytic policy gradients (SAPO-compatible)
3. P(F)-based reward gives 10x+ sample efficiency vs standard RL reward design
4. First differentiable surgical debulking simulation with policy learning

### Target Venues

| Venue | Deadline | Fit |
|---|---|---|
| CoRL 2026 | ~Jun 2026 | Best fit: robot learning + diff sim |
| ICRA 2027 | ~Sep 2026 | Needs real robot experiment |
| RSS 2026 | ~Feb 2026 | Methods focus, tight deadline |
| RA-L | Rolling | Good for incremental contribution |

## 9. Key References

| Paper | Link / Venue |
|---|---|
| SAPO / Rewarped (ICLR 2025) | arxiv.org/abs/2412.12089 |
| SHAC (ICLR 2022) | short-horizon-actor-critic.github.io |
| Stress-Guided RL (xArm + Tofu) | arxiv.org/abs/2510.25405 |
| HIL-SERL (Science Robotics 2024) | hil-serl.github.io |
| SRT-H Autonomous Surgery (Sci Rob) | doi:10.1126/scirobotics.adt5254 |
| PlasticineLab (ICLR 2021) | arxiv.org/abs/2104.03311 |
| DiffTaichi (ICLR 2020) | arxiv.org/abs/1910.00935 |
| Brax (NeurIPS 2021) | arxiv.org/abs/2106.13281 |
| Genesis (2024) | github.com/Genesis-Embodied-AI/Genesis |
| Newton (NVIDIA 2025) | developer.nvidia.com/blog/newton |
| Contact-Implicit MPC (TRO 2024) | msl.stanford.edu/papers/le_cleach_fast_2024 |
| DiffTactile (ICLR 2024) | difftactile.github.io |
| Grasp D (Diff Contact Sim) | graspd.github.io |
| PODS (Policy Opt via Diff Sim) | ICML 2021 |